

original dimension, which is essential for photolithography. As an example, polyethylene naphthalate (PEN) satisfies all requirements and tolerates temperatures as high as 160°C.

More recently, a number of electronic devices and circuits have been demonstrated, utilising paper as a substrate and/or as a gate dielectric. Such approaches lead to electronic devices with the potential for en masse integration at low cost, which are also disposable and fully recyclable.

Base material

A polymer film provides the foundation as a base material for flexible electronics. Under normal circumstances, the flex circuit base material provides most primary physical and electrical properties of the flexible circuit. In the case of adhesive-less circuit constructions, the base material provides all of the characteristic properties.

While a wide range of thickness is possible, most flexible films are provided in a narrow range of relatively thin dimensions from 12µm to 125µm (1/2 mil to 5 mils). But thinner and thicker materials are possible. Thinner materials are, of course, more flexible. For most materials, stiffness increase is proportional to the cube of thickness. This means that, if the thickness is doubled, the material becomes eight times stiffer and will only deflect 1/8 as much under the same load.

There are a number of materials used as base films, including polyester (PET), polyimide (PI), polyethylene naphthalene (PEN) and polyetherimide (PEI). These, along with various fluoropolymers (FEPs) and copolymers polyimide films, are most prevalent owing to their advantageous electrical, mechanical, chemical and thermal properties.

Bonding adhesive

Adhesives are used as the bonding medium for creating a laminate. When it comes to temperature resistance, the adhesive is also typically a performance-limiting element of a laminate, especially when polyimide is the base

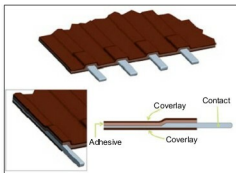


Fig. 3: Sculptured flex circuit

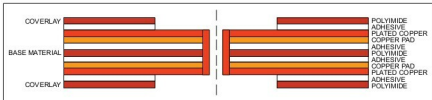


Fig. 4: Double-sided flex circuits

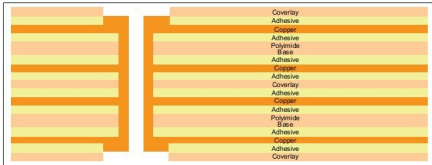


Fig. 5: Multilayer flex circuits

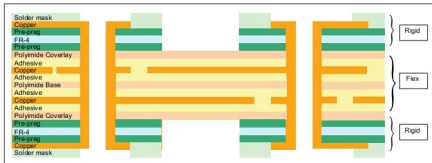


Fig. 6: Standard stack-up of a 4-layer rigid-flex circuit

material. Because of the earlier difficulties associated with polyimide adhesives, many polyimide flex circuits presently employ adhesive systems of different polymer families. However, some newer thermoplastic polyimide adhesives are making important inroads.

As with the base films, adhesives come in different thicknesses. Thick-

ness selection is typically a function of the application.

Metal foil

A metal foil is most commonly used as the conductive element of a flexible laminate. A metal foil is the material from which the circuit paths are normally etched. Copper's excellent balance of cost and physical and electrical performance attributes make it an excellent choice.

There are actually many different types of copper foil. In certain non-standard cases, the circuit manufacturer may be called upon to create a specialty laminate by using a specified alternative metal foil, such as a special copper alloy or other metal foil in the construction.

Types of flexible circuits

There are a few basic constructions of